



2022 HIGHER SCHOOL CERTIFICATE TRIAL EXAMINATION

Mathematics Extension 2

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks:
100

Section I – 10 marks (pages 2 – 4)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 5 – 11)

- Attempt Questions 11 – 16
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

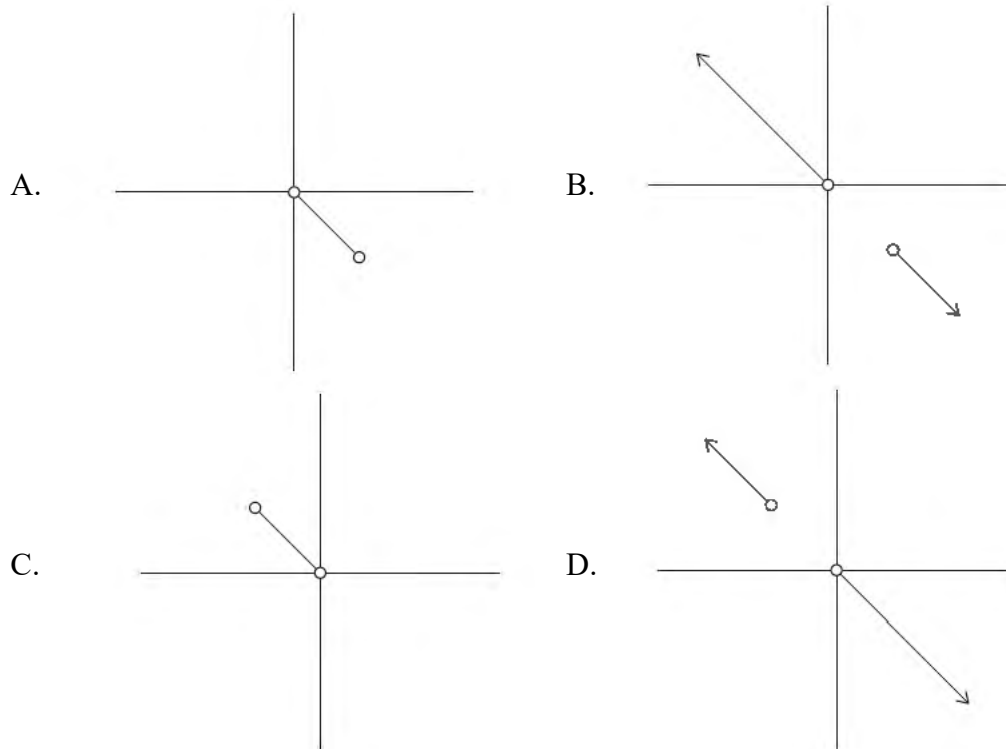
Use the multiple-choice answer sheet for Questions 1 – 10.

- 1 What is the length of the vector $-2\hat{i} - 3\hat{j} + 6\hat{k}$?
- A. 1
- B. 7
- C. 11
- D. 49
- 2 What is the converse of the following statement?
- “If a point is in the first quadrant, then its coordinates are positive”.*
- A. If a point is not in the first quadrant, then its coordinates are positive.
- B. If a point is not in the first quadrant, then its coordinates are not positive.
- C. If the coordinates of a point are positive, then the point is in the first quadrant.
- D. If the coordinates of a point are not positive, then the point is not in the first quadrant.
- 3 Multiplying a non-zero complex number by $\frac{1+i}{1-i}$ results in a rotation about the origin on an Argand diagram. What is the rotation?
- A. Clockwise by $\frac{\pi}{4}$ radians.
- B. Clockwise by $\frac{\pi}{2}$ radians.
- C. Anticlockwise by $\frac{\pi}{4}$ radians.
- D. Anticlockwise by $\frac{\pi}{2}$ radians.

- 4 Which expression is equal to $\int \cot x \, dx$?
- A. $\ln|\sin x| + c$
- B. $\frac{\cot^2 x}{2} + c$
- C. $\operatorname{cosec}^2 x + c$
- D. $\ln|\operatorname{cosec} x| + \cot x + c$
- 5 What is the angle, to the nearest degrees, between the vectors $\underline{i} + 2\underline{j} - 3\underline{k}$ and $-2\underline{i} - 4\underline{j} + 6\underline{k}$?
- A. 0
- B. 119
- C. 136
- D. 180
- 6 Which expression is equal to $\int \frac{1}{\sqrt{-x^2 + 6x - 5}} \, dx$?
- A. $\sin^{-1}\left(\frac{x-3}{2}\right) + C$
- B. $\cos^{-1}\left(\frac{x-3}{2}\right) + C$
- C. $\ln\left|x-3+\sqrt{(x-3)^2+4}\right| + C$
- D. $\ln\left|x-3+\sqrt{(x-3)^2-4}\right| + C$
- 7 Consider the statement:
- “If my computer is not working, then I cannot finish my homework”.*
- What is the negation of the above statement?
- A. My computer is working, and I can finish my homework.
- B. My computer is working, but I cannot finish my homework.
- C. My computer is not working, and I can finish my homework.
- D. My computer is not working, but I cannot finish my homework.

- 8 Which diagram best represents the solutions to the equation

$$\arg\left(\frac{z}{z-1+i}\right) = \pi ?$$



- 9 A particle is moving in simple harmonic motion with period 3 and amplitude 2. Which is a possible expression for the velocity, v , of the particle?

- A. $v = \frac{4\pi}{3} \cos\left(\frac{2\pi}{3}t\right)$
- B. $v = 2\cos\left(\frac{2\pi}{3}t\right)$
- C. $v = \frac{2\pi}{3} \cos\left(\frac{\pi}{3}t\right)$
- D. $v = 2\cos\left(\frac{\pi}{3}t\right)$

- 10 Which of the following statements is **false**?

- A. $\forall x \in \mathbb{R}, x^2 \geq 0$
- B. $\forall n \in \mathbb{Z}, \exists m \in \mathbb{Z}, m = n + 5$
- C. $\exists a \in \mathbb{R}, \forall x \in \mathbb{R}, ax = x$
- D. $\exists m \in \mathbb{Z}, \forall n \in \mathbb{Z}, m = n + 5$

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section

Answer each question on the appropriate pages of the answer booklet. Extra paper is available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (16 marks) Use the pages labelled Question 11 in the answer booklet.

(a) Consider the complex numbers $w = -2 + 3i$ and $z = 1 + i$.

(i) Evaluate $\frac{1}{|w|}$. 1

(ii) Express $w\bar{z}$ in the form $a + bi$, where a and b are real numbers. 2

(b) (i) Express $z = \sqrt{2} - \sqrt{2}i$ in modulus-argument form. 2

(ii) Hence find z^{21} in the form $a + bi$, where a and b are real numbers. 2

(c) Use integration by parts to find $\int x^3 \log_e x \, dx$. 2

(d) (i) Find the real numbers a and b , such that $\frac{5x^2 - 3x + 13}{(x-1)(x^2 + 4)} \equiv \frac{a}{x-1} + \frac{bx-1}{x^2 + 4}$. 2

(ii) Hence, find $\int \frac{5x^2 - 3x + 13}{(x-1)(x^2 + 4)} dx$. 2

(e) Using the substitution $t = \tan \frac{x}{2}$, or otherwise, calculate 3

$$\int_0^{\frac{\pi}{2}} \frac{dx}{5 + 3 \sin x + 4 \cos x}.$$

End of Question 11

Question 12 (16 marks) Use the pages labelled Question 12 in the answer booklet.

(a) Consider the vectors $\underline{a} = \begin{bmatrix} -1 \\ -2 \\ 4 \end{bmatrix}$ and $\underline{b} = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}$.

(i) Find $\underline{a} \cdot \underline{b}$. **1**

(ii) Find the vector projection of $\underline{a}$ onto $\underline{b}$. **2**

(b) Consider two points $P(-1, 3, 1)$ and $Q(2, 4, 5)$ in three-dimensional space.

(i) Find the midpoint of PQ . **1**

(ii) If P and Q are the endpoints of the diameter of a sphere, find a vector equation of the sphere. **2**

(c) Find the Cartesian equation of the line $\underline{r} = 3\underline{i} - 4\underline{j} + \lambda(2\underline{i} + \underline{j})$. **2**

(d) Consider the following proposition:

“Suppose $x, y \in \mathbb{R}$. If $y^3 + yx^2 \leq x^3 + xy^2$, then $y \leq x$.”

(i) State the contrapositive of this proposition. **1**

(ii) Hence prove the above proposition is true. **2**

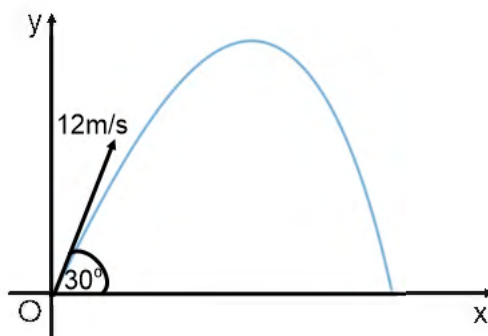
Question 12 continues on page 7

Question 12 (continued)

- (e) A projectile is fired from the origin O with initial velocity 12 ms^{-1} at an angle of 30° to the horizontal and experiences air resistance proportional to the velocity in both the x and y directions. The equations of motion are given by

$$\begin{aligned}\ddot{x} &= -k\dot{x} & \ddot{y} &= -10 - k\dot{y} \\ \dot{x} &= 6\sqrt{3}e^{-kt} & \dot{y} &= \frac{1}{k}[(10 + 6k)e^{-kt} - 10] \\ x &= \frac{6\sqrt{3}}{k}(1 - e^{-kt}) & y &= \frac{10 + 6k}{k^2}(1 - e^{-kt}) - \frac{10}{k}t\end{aligned}$$

where k is a positive constant. (Do NOT prove these results.)



- (i) Show that the Cartesian equation of the flight path is given by 2

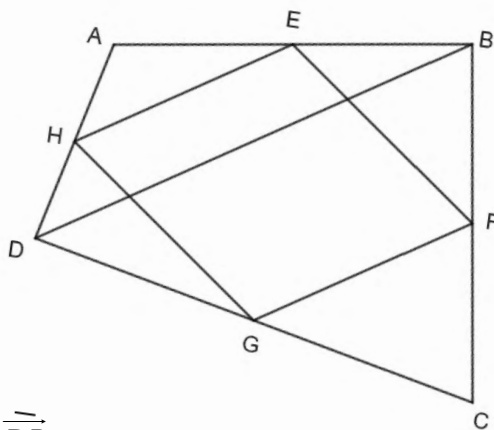
$$y = \frac{5 + 3k}{3k\sqrt{3}}x + \frac{10}{k^2} \ln \left(1 - \frac{kx}{6\sqrt{3}} \right)$$

- (ii) Given that $k = 0.1$, find the maximum height reached in metres to 2 decimal places. 3

End of Question 12

Question 13 (15 marks) Use the pages labelled Question 13 in the answer booklet.

- (a) Varignon's theorem states that the midpoints of any quadrilateral join to form a parallelogram. Let $ABCD$ be a quadrilateral and $EFGH$ be the quadrilateral obtained by joining the midpoints of the edges of $ABCD$.



- (i) Show that $\overrightarrow{HE} = \frac{1}{2} \overrightarrow{DB}$. 2
- (ii) Hence prove Varignon's theorem, i.e. show that $EFGH$ is a parallelogram. 1
- (b) Let $1, \omega, \omega^2$ be the cube roots of 1 and $\omega \neq 1$.
- (i) State the values of ω^3 and $1 + \omega + \omega^2$. 1
- (ii) Hence evaluate $(1 - \omega)(1 - \omega^2)(1 - \omega^4)(1 - \omega^5)(1 - \omega^7)(1 - \omega^8)$. 2
- (c) (i) Graph the locus of z if $|z - 2i| = 1$. 2
- (ii) Hence, or otherwise, find the maximum and minimum values of the argument of z . 2
- (d) Consider the geometric series

$$S_{2n} = 1 - h + h^2 - \dots + h^{2n}, \text{ where } 0 < h < 1.$$
- (i) Show that $S_{2n-1} < \frac{1}{1+h} < S_{2n}$. 2
- (ii) Integrate the previous result between $h = 0$ and $h = x$, where $0 < x < 1$, and hence write down a polynomial inequality for $\log_e(1 + x)$. 2
- (iii) Use $n = 3$ to estimate the value of $\log_e \frac{5}{4}$ to 4 significant figures. Show full working. 1

End of Question 13

Question 14 (14 marks) Use the pages labelled Question 14 in the answer booklet.

- (a) Let $P(z) = z^8 - \frac{5}{2}z^4 + 1$. The complex number w is a root of $P(z) = 0$.
- (i) Show that iw and $\frac{1}{w}$ are also roots of $P(z) = 0$. 1
- (ii) Hence, or otherwise, find all roots of $P(z) = 0$. 2

- (b) Find a unit vector that is perpendicular to both $\begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 4 \\ -1 \end{bmatrix}$. 3

- (c) A ball of mass 2 kilograms is thrown vertically upward from the origin with an initial speed of 8 metres per second. The ball is subject to a downward gravitational force of 20 newtons and an air resistance of $\frac{v^2}{5}$ newtons in the opposite direction to the velocity, v metres per second.

Hence, until the ball reaches its highest point, the equation of motion is

$$\ddot{y} = -\frac{v^2}{10} - 10, \quad (\text{Do NOT prove this.})$$

where y metres is its height.

- (i) Show that, while the ball is rising, $v^2 = 164e^{-\frac{y}{5}} - 100$. 2
- (ii) Hence find the maximum height reached. 1
- (iii) Find how long the ball takes to reach this maximum height. 2
- (iv) Write down the equation of motion for the downwards journey. 1
- (v) How fast is the ball travelling when it returns to the origin? 2

End of Question 14

Question 15 (15 marks) Use the pages labelled Question 15 in the answer booklet.

(a) For $n = 0, 1, 2, \dots$, let $I_n = \int_0^{\frac{\pi}{4}} \tan^n \theta \, d\theta$.

(i) Show that $I_1 = \frac{1}{2} \ln 2$. 1

(ii) Show that for $n \geq 2$, $I_n + I_{n-2} = \frac{1}{n-1}$. 2

(iii) For $n \geq 2$, explain why $I_n < I_{n-2}$, and deduce that 3

$$\frac{1}{2(n+1)} < I_n < \frac{1}{2(n-1)}.$$

(iv) By using the recurrence relation of part (ii), find I_5 and deduce that 2

$$\frac{2}{3} < \ln 2 < \frac{3}{4}.$$

(b) A particle moves in a straight line. Its displacement, x metres, after t seconds is given by

$$x = \sqrt{2} \sin\left(3t - \frac{\pi}{4}\right) + 3\sin(3t)$$

(i) Prove that the particle is moving in simple harmonic motion. 2

(ii) Find the amplitude and period of the motion. 3

(iii) When is the first time the particle reaches the origin? 2

End of Question 15

Question 16 (14 marks) Use the pages labelled Question 16 in the answer booklet.

(a) Find $\int \frac{\sqrt{25x^2 - 4}}{x} dx$. **3**

(b) Let $\underline{m}$ and $\underline{n}$ be two vectors such that $\underline{m} = a\underline{i} + b\underline{j} + c\underline{k}$ and $\underline{n} = \underline{i} + \underline{j} + \underline{k}$. **2**
If $a + b + c = 9$, prove $a^2 + b^2 + c^2 \geq 27$, where $a, b, c \in \mathbb{R}$.

(c) If $a, b, c > 0$, prove that $\frac{a}{b+c} + \frac{b}{a+c} + \frac{c}{a+b} \geq \frac{3}{2}$. **3**

(d) (i) If $y = x^k + (c - x)^k$, where $c > 0, k > 0, k \neq 1$, show that y has a single stationary point between $x = 0$ and $x = c$. **2**

(ii) Show that the stationary point is a maximum if $k < 1$ and a minimum if $k > 1$. **1**

(iii) Hence show that if $a > 0, b > 0, a \neq b$, then; **3**

$$\frac{a^k + b^k}{2} < \left(\frac{a + b}{2}\right)^k \text{ if } 0 < k < 1 \text{ and } \frac{a^k + b^k}{2} > \left(\frac{a + b}{2}\right)^k \text{ if } k > 1.$$

End of paper

BAULKHAM HILLS HIGH SCHOOL

2022 YEAR 12 EXTENSION 2 TRIAL HSC SOLUTIONS

	Solution	Marks
Section I		1
1. B	$\sqrt{(-2)^2 + (-3)^2 + 6^2} = 7$	1
2. C	Converse of $P \rightarrow Q$ is $Q \rightarrow P$	1
3. D	$\frac{1+i}{1-i} \times \frac{1+i}{1+i}$ $= \frac{1+2i+i^2}{1-i^2}$ $= \frac{2i}{2}$ $= i$ $\therefore$ An anti-clockwise rotation by $\frac{\pi}{2}$.	1
4. A	$\int \cot x \, dx$ $= \int \frac{\cos x}{\sin x} \, dx$ $= \ln \sin x + c$	1
5. D	$\cos \theta = \frac{\underline{u} \cdot \underline{v}}{ \underline{u} \underline{v} }$ $\cos \theta = \frac{1 \times (-2) + 2 \times (-4) + (-3) \times 6}{\sqrt{1^2 + 2^2 + (-3)^2} \times \sqrt{(-2)^2 + (-4)^2 + 6^2}}$ $= \frac{-28}{\sqrt{14} \times \sqrt{56}}$ $= -1$ $\theta = 180^\circ$	1
6. A	$-x^2 + 6x - 5 = -(x^2 - 6x + 5)$ $= -[(x-3)^2 - 4]$ $= 4 - (x-3)^2$ $\therefore \int \frac{1}{\sqrt{-x^2 + 6x - 5}} \, dx$ $= \int \frac{1}{\sqrt{2^2 - (x-3)^2}} \, dx$ $= \sin^{-1} \left(\frac{x-3}{2} \right) + C$	1

7. C	$\neg(P \rightarrow Q)$ $= \neg(\neg P \vee Q)$ $= \neg\neg P \wedge \neg Q$ $= P \wedge \neg Q$	1
8. A	$\arg\left(\frac{z}{z-1+i}\right) = \pi$ $\arg(z) - \arg(z - (1-i)) = \pi$ $\therefore$ The difference in angle is π . $\therefore z$ lies between (0,0) and (1,-1).	1
9. A	Consider answer A: $v = \frac{4\pi}{3} \cos\left(\frac{2\pi}{3}t\right)$ $x = 2\pi \sin\left(\frac{2\pi}{3}t\right) + C$ $\therefore$ Amplitude is 2 and Period is $\frac{2\pi}{\left(\frac{2\pi}{3}\right)} = 3$. (B, C and D does not work).	1
10. D	A) For every real number x , $x^2 \geq 0$. True B) For all integers n , there exist an integer m such that $m = n + 5$. True C) There exists a real number a for which $ax=x$ for every real number x . True D) There exists an integer m , for all integers n such that $m = n + 5$. False	1

Question	Solution	Mark	Comments
Section II			
11(a)(i)	$\frac{1}{ w } = \frac{1}{\sqrt{(-2)^2 + 3^2}} = \frac{1}{\sqrt{13}}$	1	1Mk: Provides correct solution.
11(a)(ii)	$w\bar{z} = (-2+3i)(1-i)$ $= -2 + 2i + 3i - 3i^2$ $= 1 + 5i$	2	2Mk: Provides correct solution. 1Mk: Obtains correct $\bar{z}$.
11(b)(i)	$z = \sqrt{2} - \sqrt{2}i$ $ z = \sqrt{2+2} = 2$ $\& \arg(z) = \tan^{-1}\left(\frac{-\sqrt{2}}{\sqrt{2}}\right) = -\frac{\pi}{4}$ $\therefore z = 2\left(\cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right)\right)$	2	2Mk: Provides correct solution. 1Mk: Obtains correct modulus or argument.

11(b)(ii)	$z^{21} = 2^{21} \left(\cos \left(21 \times -\frac{\pi}{4} \right) + i \sin \left(21 \times -\frac{\pi}{4} \right) \right)$ (De Moivre's Theorem) $= 2^{21} \left(\cos \left(-\frac{21\pi}{4} \right) + i \sin \left(-\frac{21\pi}{4} \right) \right)$ $= 2^{21} \left(\cos \left(\frac{3\pi}{4} \right) + i \sin \left(\frac{3\pi}{4} \right) \right)$ $= 2097152 \left(-\frac{1}{\sqrt{2}} + i \left(-\frac{1}{\sqrt{2}} \right) \right)$ $= -1048576\sqrt{2} + 1048576\sqrt{2}i$	2	2Mk: Provides correct solution. 1Mk: Uses De Moivre's Theorem correctly.
11(c)	$\int x^3 \ln x \, dx$ $u = \ln x \quad v = \frac{1}{4}x^4$ $du = \frac{dx}{x} \quad dv = x^3$ $= \left(\frac{1}{4}x^4 \right) (\ln x) - \int \frac{1}{4}x^3 \, dx$ $= \frac{1}{4}x^4 \ln x - \frac{x^4}{16} + C$	2	2Mk: Provides correct solution. 1Mk: Attempts to use integration by parts.
11(d)(i)	$\frac{5x^2-3x+13}{(x-1)(x^2+4)} \equiv \frac{a}{x-1} + \frac{bx-1}{x^2+4}$ $= \frac{a}{x-1} + \frac{bx-1}{x^2+4}$ $= \frac{a(x^2+4) + (bx-1)(x-1)}{(x-1)(x^2+4)}$ $\therefore a(x^2+4) + (bx-1)(x-1) = 5x^2 - 3x + 13$ When $x = 1$, $5a = 15$ $a = 3$ Equating the coefficient of x^2 gives $a + b = 5$ $b = 2$ $\therefore a = 3, b = 2$	2	2Mk: Provides correct solution. 1Mk: Obtains a or b correctly.
11(d)(ii)	$\int \frac{5x^2-3x+13}{(x-1)(x^2+4)} \, dx$ $= \int \frac{3}{x-1} + \frac{2x-1}{x^2+4} \, dx$ $= 3 \ln x-1 + \int \frac{2x \, dx}{x^2+4} - \int \frac{dx}{x^2+4}$ $= 3 \ln x-1 + \ln(x^2+4) - \frac{1}{2} \tan^{-1} \frac{x}{2} + C$	2	2Mk: Provides correct solution. 1Mk: Correctly integrates for a $\ln$ or inverse $\tan$.

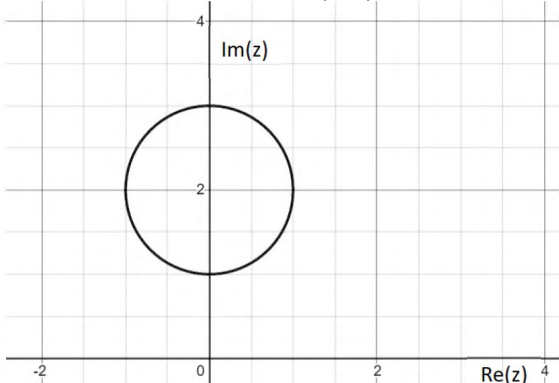
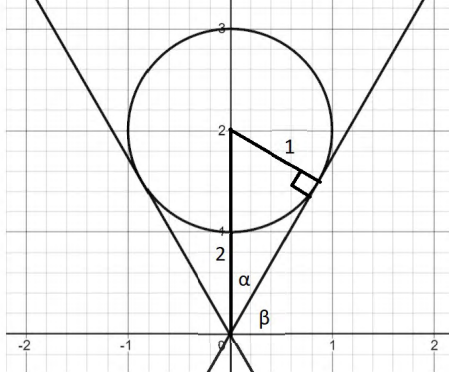
11(e)	$\int_0^{\frac{\pi}{2}} \frac{dx}{5 + 3 \sin x + 4 \cos x}$ Let $t = \tan \frac{x}{2}$ $\therefore \frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$ $= \frac{1}{2} \left(\tan^2 \frac{x}{2} + 1 \right)$ $\frac{dt}{dx} = \frac{1}{2} (t^2 + 1)$ $\therefore \frac{2dt}{1+t^2} = dx$ When $x = \frac{\pi}{2}$, $t = 1$ When $x = 0$, $t = 0$ $\therefore = \int_0^1 \frac{2dt}{(1+t^2) \left[5 + 3 \left(\frac{2t}{1+t^2} \right) + 4 \left(\frac{1-t^2}{1+t^2} \right) \right]}$ $= \int_0^1 \frac{2dt}{5 + 5t^2 + 6t + 4 - 4t^2}$ $= \int_0^1 \frac{2dt}{(t+3)^2}$ $= -2 \left[\frac{1}{t+3} \right]_0^1$ $= -2 \left(\frac{1}{4} - \frac{1}{3} \right)$ $= \frac{1}{6}$	3	3Mk: Provides correct solution. 2Mk: Integrates correctly. 1Mk: Obtains correct integrand in t.
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12(a)(i)	$\underline{a} \cdot \underline{b} = (-1)(3) + (-2)(-1) + (4)(2)$ $= -3 + 2 + 8$ $= 7$	1	1Mk: Provides correct solution.
12(a)(ii)	$ \underline{b} = \sqrt{3^2 + (-1)^2 + 2^2}$ $= \sqrt{9 + 1 + 4}$ $= \sqrt{14}$ $proj_{\underline{b}} \underline{a} = \frac{\underline{a} \cdot \underline{b}}{ \underline{b} \underline{b} } \underline{b}$ $= \frac{7}{\sqrt{14} \times \sqrt{14}} \underline{b}$ $= \frac{1}{2} (3\underline{i} - \underline{j} + 2\underline{k})$ $= \frac{3}{2} \underline{i} - \frac{1}{2} \underline{j} + \underline{k}$	2	2Mk: Provides correct solution. 1Mk: Obtains $ \underline{b} ^2$ correctly.
12(b)(i)	$\text{Midpoint} = \left(\frac{-1+2}{2}, \frac{3+4}{2}, \frac{1+5}{2} \right)$ $= \left(\frac{1}{2}, \frac{7}{2}, 3 \right)$	1	1Mk: Provides correct solution.
12(b)(ii)	Radius is the distance from midpoint to P: $= \sqrt{\left(\frac{1}{2} + 1 \right)^2 + \left(\frac{7}{2} - 3 \right)^2 + (3 - 1)^2}$ $= \sqrt{\frac{13}{2}}$ Equation of the sphere: $\left \underline{r} - \begin{bmatrix} 0.5 \\ 3.5 \\ 3 \end{bmatrix} \right = \sqrt{\frac{13}{2}}$ Must be a vector equation.	2	2Mk: Provides correct solution. 1Mk: Obtains correct radius.
12(c)	$\underline{r} = 3\underline{i} - 4\underline{j} + \lambda(2\underline{i} + \underline{j})$ $\therefore x = 3 + 2\lambda$ $y = -4 + \lambda$ $y + 4 = \lambda$ $\therefore x = 3 + 2(y + 4)$ $\therefore x - 2y - 11 = 0$ or $y = \frac{x}{2} - \frac{11}{2}$	2	2Mk: Provides correct solution. 1Mk: Attempts to solve simultaneously or finds the slope of the line.

12(d)(i)	Contrapositive is: If $y > x$, then $y^3 + yx^2 > x^3 + xy^2$	1	1Mk: Provides correct solution.
12(d)(ii)	Proving the contrapositive is true. I.e. if $y > x$ then $y^3 + yx^2 > x^3 + xy^2$. $y > x$ $y(x^2 + y^2) > x(x^2 + y^2)$ (since $x^2 + y^2 > 0$) $y^3 + yx^2 > x^3 + xy^2$ as required. ∴ The contrapositive is true. By contraposition ∴ If $y \leq x$ then $y^3 + yx^2 \leq x^3 + xy^2$	2	2Mk: Provides correct solution. 1Mk: Multiply both sides by $(x^2 + y^2)$ or correct algebraic method towards the inequality.
12(e)(i)	$\frac{kx}{6\sqrt{3}} = 1 - e^{-kt}$ $e^{-kt} = 1 - \frac{6\sqrt{3}}{kx}$ $-kt = \ln \left(1 - \frac{6\sqrt{3}}{kx} \right)$ $t = -\frac{1}{k} \ln \left(1 - \frac{6\sqrt{3}}{kx} \right)$ Substitute t into y : $y = \frac{10 + 6k}{10 + 6k} \cdot \frac{k^2}{kx} - \frac{6\sqrt{3}}{kx} \cdot \frac{k}{10} \left[\frac{1}{1} \ln \left(1 - \frac{6\sqrt{3}}{kx} \right) \right]$ $= \frac{10 + 6k}{10 + 6k} \cdot \frac{k^2}{kx} + \frac{6\sqrt{3}}{10} \ln \left(1 - \frac{6\sqrt{3}}{kx} \right)$ $= \frac{5 + 3k}{5 + 3k} \cdot \frac{k^2}{x} + \frac{6\sqrt{3}}{10} \ln \left(1 - \frac{6\sqrt{3}}{kx} \right)$ as required.	2	2Mk: Provides correct proof. 1Mk: Obtains correct t .

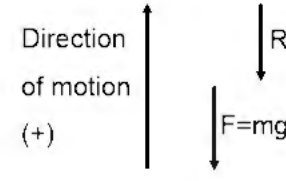
12(e)(ii)	Maximum height occurs when $\dot{y} = 0$: $0 = \frac{1}{k}[(10 + 6k)e^{-kt} - 10]$ $(10 + 6k)e^{-kt} = 10$ $e^{-kt} = \frac{10}{10 + 6k}$ $e^{-kt} = 1 + 0.6k$ $kt = \ln(1 + 0.6k)$ $t = \frac{1}{k} \ln(1 + 0.6k)$ When $k = 0.1$, $t = \frac{1}{0.1} \ln(1 + 0.6 \times 0.1)$ $t = 0.582689...$ Sub t into y: $y = \frac{10 + 6k}{k^2}(1 - e^{-kt}) - \frac{10}{k}t$ $y = \frac{10 + 6(0.1)}{0.1^2}(1 - e^{-(0.1)(0.582689...)}) - \frac{10}{0.1}(0.582689...)$ $y = 1.73109...$ $y = 1.73\text{m}$	3	3Mk: Provides correct solution. 2Mk: Obtains correct value for t or correct expression for t. 1Mk: Attempts to find t when $\dot{y} = 0$.
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13(a)(i)	$\overrightarrow{HE} = \overrightarrow{HA} + \overrightarrow{AE}$ $= \frac{1}{2}(\overrightarrow{DA} + \overrightarrow{AB})$ $= \frac{1}{2}(\overrightarrow{DB}) \quad \text{as required.}$	2	2Mk: Provides correct solution. 1Mk: Correctly applies one addition rule.
13(a)(ii)	Similarly $\overrightarrow{GF} = \frac{1}{2}(\overrightarrow{DB})$ $= \overrightarrow{HE}$ $\therefore EFGH$ is a parallelogram. (One pair of opposite sides of a quadrilateral parallel and equal in length).	1	1Mk: Provides correct explanation.

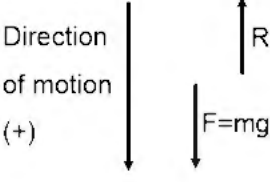
13(b)(i)	ω is the cube root of 1. $\therefore \omega^3 = 1$ $\omega^3 - 1 = 0$ $(\omega - 1)(\omega^2 + \omega + 1) = 0$ $\therefore \omega^2 + \omega + 1 = 0$	1	1Mk: Provides correct answers.
13(b)(ii)	$(1 - \omega)(1 - \omega^2)(1 - \omega^4)(1 - \omega^5)(1 - \omega^7)(1 - \omega^8)$ $= (1 - \omega)(1 - \omega^2)(1 - \omega^1)(1 - \omega^2)(1 - \omega^1)(1 - \omega^2)$ $= (1 - \omega)^3(1 - \omega^2)^3$ $= ((1 - \omega)(1 - \omega^2))^3$ $= (1 - \omega^2 - \omega + \omega^3)^3$ $= (1 + 1 + \omega^3)^3 \quad [-\omega^2 - \omega = 1 \text{ from (i)}]$ $= (1 + 1 + 1)^3 \quad [\omega^3 = 1 \text{ from (i)}]$ $= 3^3$ $= 27$	2	2Mk: Provides correct solution. 1Mk: Substitutes $\omega^3 = 1$.
13(c)(i)	It is a circle with centre (0,2) and radius 1. 	2	2Mk: Provides correct sketch. 1Mk: Gives correct shape.
13(c)(ii)	 For the minimum value of $\arg(z)$, it must be a tangent to the circle. Use trigonometry to solve for α and β. We get $\alpha = \frac{\pi}{6}$ and $\beta = \frac{\pi}{3}$. $\therefore \text{Minimum } \arg(z) = \frac{\pi}{3}$. Similarly, maximum $\arg(z) = \frac{2\pi}{3}$.	2	2Mk: Provides correct solution. 1Mk: Identify the minimum $\arg(z)$ is a tangent and finds the angle.

13(d)(i)	$S_{2n-1} = 1 - h + h^2 - h^3 + \dots - h^{2n-1}$ $= \frac{1 - (-h)^{2n}}{1 - (-h)} \quad (\text{sum of GP})$ $= \frac{1 - h^{2n}}{1 + h}$ $< \frac{1}{1 + h} \quad (\text{greater numerator})$ $\& S_{2n} = 1 - h + h^2 - h^3 + \dots + h^{2n}$ $= \frac{1 - (-h)^{2n+1}}{1 - (-h)} \quad (\text{sum of GP})$ $= \frac{1 + h^{2n+1}}{1 + h}$ $> \frac{1}{1 + h} \quad (\text{smaller numerator})$ $\therefore S_{2n-1} < \frac{1}{1 + h} < S_{2n} \quad \text{as required.}$	2	2Mk: Provides correct solution. 1Mk: Correctly proves one inequality.
13(d)(ii)	$\int_0^x S_{2n-1} dh < \int_0^x \frac{1}{1+h} dh < \int_0^x S_{2n} dh$ $\left[h - \frac{h^2}{2} + \dots - \frac{h^{2n}}{2n} \right]_0^x < [\ln(1+h)]_0^x < \left[h - \frac{h^2}{2} + \dots + \frac{h^{2n+1}}{2n+1} \right]_0^x$ $x - \frac{x^2}{2} + \dots - \frac{x^{2n}}{2n} < \log_e(1+x) < x - \frac{x^2}{2} + \dots + \frac{x^{2n+1}}{2n+1}$	2	2Mk: Provides correct solution. 1Mk: Integrates one side correctly, including obtaining $\log_e(1+x)$.
13(d)(iii)	Use $n = 3$ and $x = 0.25$ $0.25 - \frac{(0.25)^2}{2} + \frac{(0.25)^3}{3} - \frac{(0.25)^4}{4} + \frac{(0.25)^5}{5} - \frac{(0.25)^6}{6} < \ln(1+0.25) <$ $0.25 - \frac{(0.25)^2}{2} + \frac{(0.25)^3}{3} - \frac{(0.25)^4}{4} + \frac{(0.25)^5}{5} - \frac{(0.25)^6}{6} + \frac{(0.25)^7}{7}$ $0.223136... < \ln\left(\frac{5}{4}\right) < 0.223145...$ $\therefore \ln\left(\frac{5}{4}\right) = 0.2231 \text{ (to 4 s.f.)}$	1	1Mk: Provides correct solution.

14(a)(i)	$P(z) = z^8 - \frac{5}{2}z^4 + 1 \text{ and } w \text{ is a root of } P(z) = 0.$ Show that iw and $\frac{1}{w}$ are also roots of $P(z) = 0$. $P(iw) = (iw)^8 - \frac{5}{2}(iw)^4 + 1$ $= i^8 w^8 - \frac{5}{2}i^4 w^4 + 1$ $= w^8 - \frac{5}{2}w^4 + 1$ $= 0$ $P\left(\frac{1}{w}\right) = \left(\frac{1}{w}\right)^8 - \frac{5}{2}\left(\frac{1}{w}\right)^4 + 1$ $= \frac{1}{w^8} \left(1 - \frac{5}{2}w^4 + w^8\right)$ $= \frac{1}{w^8} \times 0$ $= 0$	1	1Mk: Correctly shows the results.
14(a)(ii)	$z^8 - \frac{5}{2}z^4 + 1 = 0$ $\therefore 2z^8 - 5z^4 + 2 = 0$ $\therefore (2z^4 - 1)(z^4 - 2) = 0$ $\therefore z^4 = \frac{1}{2} \text{ or } 2$ $\therefore \text{One root is } z = \sqrt[4]{2}$ $\therefore z = \pm\sqrt[4]{2}, z = \pm\sqrt[4]{2}i, z = \pm\frac{1}{\sqrt[4]{2}}, z = \pm\frac{1}{\sqrt[4]{2}i}$	2	2Mk: Provides correct solution. 1Mk: Obtains one correct value for z .

14(b)	Let $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ be perpendicular to $\begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 4 \\ -1 \end{bmatrix}$ $\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} \cdot \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix} = 0 \quad \therefore 3x - 2y + z = 0$ $\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 4 \\ -1 \end{bmatrix} = 0 \quad \therefore x + 4y - z = 0$ Solve simultaneously: $4x + 2y = 0$ $y = -2x$ Sub y into second equation: $x - 8x - z = 0$ $z = -7x$ $\therefore \begin{bmatrix} x \\ -2x \\ -7x \end{bmatrix} \text{ is perpendicular to the two given vectors.}$ Let $x = 1$, $\begin{bmatrix} 1 \\ -2 \\ -7 \end{bmatrix}$ is a one particular vector that is perpendicular. For unit vector: $\hat{u} = \frac{1}{\sqrt{1+4+49}} \begin{bmatrix} 1 \\ -2 \\ -7 \end{bmatrix}$ $= \frac{1}{\sqrt{54}} \begin{bmatrix} 1 \\ -2 \\ -7 \end{bmatrix}$ $= \frac{1}{3\sqrt{6}} \begin{bmatrix} 1 \\ -2 \\ -7 \end{bmatrix}$	3	3Mk: Provides correct unit vector. 2Mk: Obtains a general vector that is perpendicular to the given vectors. 1Mk: Obtains correct Cartesian equations.
14(c)	 $m\ddot{y} = -(F + R)$ $2\ddot{y} = -20 - \frac{v^2}{5}$ $\ddot{y} = -10 - \frac{v^2}{10}$		

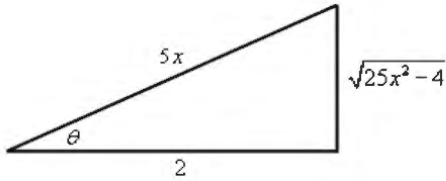
14(c)(i)	$\ddot{y} = v \frac{dv}{dy}$ $\therefore v \frac{dv}{dy} = -10 - \frac{v^2}{10}$ $\frac{dv}{dy} = \frac{-10}{v} - \frac{v}{10}$ $\frac{dy}{dv} = \frac{-10v}{v^2 + 100}$ $\therefore [y]_0^y = \int_8^v \frac{-10v}{v^2 + 100} dv$ $y = -5 \left[\ln(v^2 + 100) \right]_8^v$ $-\frac{y}{5} = \ln(v^2 + 100) - \ln 164$ $-\frac{y}{5} = \ln \left(\frac{v^2 + 100}{164} \right)$ $\frac{v^2 + 100}{164} = e^{-\frac{y}{5}}$ $\therefore v^2 = 164e^{-\frac{y}{5}} - 100 \quad \text{as required.}$	2	2Mk: Correctly shown. 1Mk: Attempts to integrate $\frac{-10v}{v^2 + 100}$.
14(c)(ii)	Max height occurs when $v = 0$. $\therefore 0 = 164e^{-\frac{y}{5}} - 100$ $100 = 164e^{-\frac{y}{5}}$ $-5 \ln \left(\frac{100}{164} \right) = y$ $\therefore y = 5 \ln(1.64) \text{ m or } 2.47 \text{ m}$	1	1Mk: Provides correct answer.
14(c)(iii)	$\ddot{y} = \frac{dv}{dt}$ $\therefore \frac{dv}{dt} = -10 - \frac{v^2}{10}$ $\frac{dt}{dv} = \frac{-10}{v^2 + 100}$ $\therefore t = \int_8^0 \frac{-10}{v^2 + 100} dv$ $= -10 \times \frac{1}{10} \left[\tan^{-1} \frac{v}{10} \right]_8^0$ $= \tan^{-1} \left[\frac{v}{10} \right]_0^8$ $= \tan^{-1} \left(\frac{4}{5} \right)$ $\therefore \text{Reaches max height after } \tan^{-1} \left(\frac{4}{5} \right) \text{ seconds.}$	2	2Mk: Provides correct solution. 1Mk: Correctly integrates $\frac{-10}{v^2 + 100}$.

14(c)(iv)	Direction of motion (+)  $m\ddot{y} = F - R$ $2\ddot{y} = 20 - \frac{v^2}{5}$ $\therefore \ddot{y} = 10 - \frac{v^2}{10}$	1	1Mk: Provides correct solution.
14(c)(v)	$v \frac{dv}{dy} = 10 - \frac{v^2}{10}$ $\frac{dv}{dy} = \frac{10}{v} - \frac{v}{10}$ $\frac{dy}{dv} = \frac{10v}{100 - v^2}$ $[y]_0^{5 \ln 1.64} = \int_0^v \frac{10v}{100 - v^2} dv$ $5 \ln 1.64 = -5 \left[\ln(100 - v^2) \right]_0^v$ $-\ln 1.64 = \ln(100 - v^2) - \ln 100$ $\ln(100 - v^2) = \ln \left(\frac{100}{1.64} \right)$ $v^2 = \frac{64}{1.64}$ $v = \frac{8}{\sqrt{1.64}} \quad (\text{since } v > 0)$ $v = 6.25 \text{ m/s}$ $\therefore$ Ball is travelling at 6.25 m/s when it returns to the origin.	2	2Mk: Provides correct solution. 1Mk: Integrates correctly.
15(a)(i)	$I_n = \int_0^{\frac{\pi}{4}} \tan^n \theta d\theta$ $I_1 = \int_0^{\frac{\pi}{4}} \tan \theta d\theta$ $= [-\ln(\cos \theta)]_0^{\frac{\pi}{4}}$ $= \ln[\cos(0)] - \ln \left[\cos \left(\frac{\pi}{4} \right) \right]$ $= \ln 1 - \ln \left(\frac{1}{\sqrt{2}} \right)$ $= 0 - \ln \left(2^{-\frac{1}{2}} \right)$ $= \frac{1}{2} \ln 2 \quad \text{as required.}$	1	1Mk: Provides correct proof.

15(a)(ii)	Show that for $n \geq 2$, $I_n + I_{n-2} = \frac{1}{n-1}$. $I_n = \int_0^{\frac{\pi}{4}} \tan^{n-2} \theta (\sec^2 \theta - 1) d\theta, \text{ where } n \geq 2$ $= \int_0^{\frac{\pi}{4}} \tan^{n-2} \theta \sec^2 \theta - \tan^{n-2} \theta d\theta$ $= \int_0^{\frac{\pi}{4}} \tan^{n-2} \theta \sec^2 \theta d\theta - I_{n-2}$ $\therefore I_n + I_{n-2} = \int_0^{\frac{\pi}{4}} \tan^{n-2} \theta \sec^2 \theta d\theta$ $= \left[\frac{\tan^{n-1} \theta}{n-1} \right]_0^{\frac{\pi}{4}}$ $= \frac{(\tan^{n-1} \theta)^{n-1}}{n-1} - \frac{(\tan(0))^{n-1}}{n-1}$ $= \frac{1}{n-1} - 0$ $= \frac{1}{n-1}$	2	2Mk: Provides correct proof. 1Mk: Obtains correct integral for $I_n + I_{n-2}$ or uses integration by parts correctly towards I_n or I_{n-2}.
15(a)(iii)	For $0 < \theta < \frac{\pi}{4}$, $0 < \tan \theta < 1$ (since $\tan \theta$ is monotonic increasing on the domain) $0 < \tan^2 \theta < 1$ $0 < \tan^n \theta < \tan^{n-2} \theta$ (since $\tan^{n-2} > 0$) $\therefore \int_0^{\frac{\pi}{4}} \tan^n \theta d\theta < \int_0^{\frac{\pi}{4}} \tan^{n-2} \theta d\theta$ that is, $I_n < I_{n-2}$ $\therefore 2I_n < I_{n-2} + I_n \quad (1)$ $2I_n < \frac{1}{n-1} \quad [\text{from (ii)}]$ $I_n < \frac{1}{2(n-1)}$ Replacing n by $n+2$ in (1), we have $I_{n+2} < I_n$ $I_{n+2} + I_n < 2I_n$ $\frac{1}{n+2-1} < 2I_n \quad [\text{from (ii)}]$ $\frac{1}{2(n+1)} < I_n$ Hence $\frac{1}{2(n+1)} < I_n < \frac{1}{2(n-1)}$.	3	3Mk: Provides correct proof. 2Mk: Correctly explains $I_n < I_{n-2}$ and correctly proves part of the compound inequality or proving the compound inequality. 1Mk: Sufficiently explains why $I_n < I_{n-2}$.

15(a)(iv)	$I_5 = \frac{1}{4} - I_3 \quad [\text{from (ii)}]$ $= \frac{1}{4} - \left(\frac{1}{2} - I_1 \right) \quad [\text{from (ii)}]$ $= -\frac{1}{4} + \frac{1}{2} \ln 2 \quad [\text{from (i)}]$ $= \frac{2 \ln 2 - 1}{4}$ From (iii), we have $\frac{1}{2 \times 6} < I_5 < \frac{1}{2 \times 4}$ $\frac{1}{12} < \frac{2 \ln 2 - 1}{4} < \frac{1}{8}$ $\frac{1}{3} < 2 \ln 2 - 1 < \frac{1}{2}$ $\frac{4}{3} < 2 \ln 2 < \frac{3}{2}$ $\frac{2}{3} < \ln 2 < \frac{3}{4} \quad \text{as required.}$	2	2Mk: Provides correct proof. 1Mk: Finds I_5 by using the recurrence relation.
15(b)(i)	$x = \sqrt{2} \sin \left(3t - \frac{\pi}{4} \right) + 3 \sin(3t)$ $\dot{x} = 3\sqrt{2} \cos \left(3t - \frac{\pi}{4} \right) + 9 \cos(3t)$ $\ddot{x} = -9\sqrt{2} \sin \left(3t - \frac{\pi}{4} \right) - 27 \sin(3t)$ $= -9 \left(\sqrt{2} \sin \left(3t - \frac{\pi}{4} \right) + 3 \sin(3t) \right)$ $= -3^2 x$ In the form $\ddot{x} = -n^2 x$, where $n = 3$ and centre is 0. The motion is simple harmonic.	2	2Mk: Provides correct solution. 1Mk: Attempts to find $\dot{x}$ and $\ddot{x}$.
15(b)(ii)	Period is $\frac{2\pi}{3}$. Amplitude: $x = \sqrt{2} \sin \left(3t - \frac{\pi}{4} \right) + 3 \sin(3t)$ $= \sqrt{2} \left(\sin(3t) \cos \left(\frac{\pi}{4} \right) - \sin \left(\frac{\pi}{4} \right) \cos(3t) \right) + 3 \sin(3t)$ $= \sqrt{2} \left(\frac{1}{\sqrt{2}} \sin(3t) - \frac{1}{\sqrt{2}} \cos(3t) \right) + 3 \sin(3t)$ $= -\cos(3t) + (1+3) \sin(3t)$ $= 4 \sin(3t) - \cos(3t)$ This can be expressed in the form $A \sin(3t - \alpha)$. $A \sin(3t - \alpha) = A \sin(3t) \cos \alpha - A \sin \alpha \cos(3t)$ $\therefore A \cos \alpha = 4$ $A \sin \alpha = 1$ $\therefore A = \sqrt{4^2 + 1^2}$ $= \sqrt{17}$ $\therefore$ Amplitude is $\sqrt{17}$	3	3Mk: Provides correct period and amplitude. 2Mk: Obtains correct period and significant progress towards amplitude or correct amplitude. 1Mk: Obtains correct period or significant progress towards amplitude.

15(b)(iii)	From (ii): $A \cos \alpha = 4$ $A \sin \alpha = 1$ $\tan \alpha = \frac{1}{4}$ $\alpha = \tan^{-1}\left(\frac{1}{4}\right)$ $\therefore \sqrt{2} \sin\left(3t - \frac{\pi}{4}\right) + 3 \sin(3t) = \sqrt{17} \sin\left(3t - \tan^{-1}\left(\frac{1}{4}\right)\right)$ Reaches the origin when $x = 0$. $\sin\left(3t - \tan^{-1}\left(\frac{1}{4}\right)\right) = 0$ $3t - \tan^{-1}\left(\frac{1}{4}\right) = 0, \pi, 2\pi, \dots$ $3t = 0 + \tan^{-1}\left(\frac{1}{4}\right), \pi + \tan^{-1}\left(\frac{1}{4}\right), \dots$ $t = \frac{\tan^{-1}\left(\frac{1}{4}\right)}{3}, \frac{\pi + \tan^{-1}\left(\frac{1}{4}\right)}{3}, \dots$ $t = 0.081659\dots, 1.128857\dots, \dots \quad \text{Using radians mode.}$ $\therefore t = 0.0817\text{s} \text{ is the first time the particle is at the origin.}$	2	2Mk: Provides correct solution. 1Mk: Obtains correct value for α or uses $x = 0$ to find an expression for time.
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16(a)	$x = \frac{2}{5} \sec \theta \Rightarrow dx = \frac{2}{5} \sec \theta \tan \theta d\theta$ $\int \frac{\sqrt{25x^2 - 4}}{x} dx = \int \frac{2 \tan \theta}{\frac{2}{5} \sec \theta} \left(\frac{2}{5} \sec \theta \tan \theta d\theta \right)$ $= 2 \int \tan^2 \theta$ $= 2 \int \sec^2 \theta - 1 d\theta$ $= 2(\tan \theta - \theta) + C$ Since $\sec \theta = \frac{5x}{2} = \frac{\text{hyp}}{\text{adj}}$  $\therefore \tan \theta = \frac{\sqrt{25x^2 - 4}}{2}$ $\& \cos \theta = \frac{2}{5x} \Rightarrow \theta = \cos^{-1}\left(\frac{2}{5x}\right)$ $\therefore \int \frac{\sqrt{25x^2 - 4}}{x} dx = \sqrt{25x^2 - 4} - 2 \cos^{-1}\left(\frac{2}{5x}\right) + C$	3	3Mk: Provides correct solution. 2Mk: Integrates correctly to get $2(\tan \theta - \theta) + C$. 1Mk: Uses a valid substitution.
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16(b)	$\underline{m} = a\underline{i} + b\underline{j} + c\underline{k} \quad \& \quad \underline{n} = \underline{i} + \underline{j} + \underline{k}$ $\therefore \underline{m} \cdot \underline{n} \leq \underline{m} \underline{n} \quad (\text{from } \underline{m} \cdot \underline{n} = \underline{m} \underline{n} \cos \theta)$ $\therefore a + b + c \leq \sqrt{a^2 + b^2 + c^2} \times \sqrt{3}$ $9 \leq \sqrt{a^2 + b^2 + c^2} \times \sqrt{3} \quad (\text{from assumption})$ $\sqrt{27} \leq \sqrt{a^2 + b^2 + c^2}$ $a^2 + b^2 + c^2 \geq 27 \quad \text{as required.}$	2	2Mk: Provides correct proof. 1Mk: Identifies $\underline{m} \cdot \underline{n} \leq \underline{m} \underline{n} $ or obtains $a^2 + b^2 + c^2 \geq ab + bc + ca$ by algebraic method.
16(c)	$\text{LHS} = \left(\frac{a}{b+c} + 1 \right) + \left(\frac{b}{a+c} + 1 \right) + \left(\frac{c}{a+b} + 1 \right) - 3$ $= \frac{a+b+c}{b+c} + \frac{a+b+c}{a+c} + \frac{a+b+c}{a+b} - 3$ $= (a+b+c) \left(\frac{1}{b+c} + \frac{1}{a+c} + \frac{1}{a+b} \right) - 3$ $= \frac{1}{2} \times ((b+c) + (a+c) + (a+b)) \times \left(\frac{1}{b+c} + \frac{1}{a+c} + \frac{1}{a+b} \right) - 3$ $\geq \frac{1}{2} \times (3 \times \sqrt[3]{(b+c)(a+c)(a+b)}) \times \left(3 \times \sqrt[3]{\frac{1}{(b+c)(a+c)(a+b)}} \right) - 3$ (by AM/GM with $n=3$) $= \frac{9}{2} \times \sqrt[3]{1} - 3$ $= \frac{3}{2} \quad \text{as required.}$	3	3Mk: Provides correct solution. 2Mk: Shows significant progress. 1Mk: Makes a valid attempt.
16(d)(i)	$y = x^k + (c-x)^k$ $\frac{dy}{dx} = kx^{k-1} - k(c-x)^{k-1}$ stationary points occur when $\frac{dy}{dx} = 0$ $kx^{k-1} - k(c-x)^{k-1} = 0$ $x^{k-1} = (c-x)^{k-1}$ $\left(\frac{c-x}{x} \right)^{k-1} = 1$ $\frac{c-x}{x} = \pm 1$ (note: if $k-1$ is even it is possible that $c-x$ is < 0) $\frac{c-x}{x} = -1 \quad \text{or} \quad \frac{c-x}{x} = 1$ $\frac{c}{x} - 1 = -1 \quad \frac{c}{x} - 1 = 1$ $\frac{c}{x} = 0 \quad \frac{c}{x} = 2$ not possible as $c > 0$ $x = \frac{c}{2}$ $\therefore y$ has a single stationary point at $x = \frac{c}{2}$	2	2Mk: Provides correct solution by considering both cases. 1Mk: Finds the stationary value of $x = \frac{c}{2}$.

16(d)(ii)	$\frac{d^2y}{dx^2} = k(k-1)x^{k-2} + k(k-1)(c-x)^{k-2}$ when $x = \frac{c}{2}$; $\frac{d^2y}{dx^2} = k(k-1)\left(\frac{c}{2}\right)^{k-2} + k(k-1)\left(\frac{c}{2}\right)^{k-2}$ $= 2k(k-1)\left(\frac{c}{2}\right)^{k-2}$ If $0 < k < 1$ then $k(k-1) < 0$ and if $k > 1$ then $k(k-1) > 0$ Thus if $k < 1$, y has a maximum at $x = \frac{c}{2}$ and if $k > 1$, y has a minimum at $x = \frac{c}{2}$.	1	1Mk: Provides correct solution.
16(d)(iii)	when $x = \frac{c}{2}$; $y = \left(\frac{c}{2}\right)^k + \left(\frac{c}{2}\right)^k$ $= 2\left(\frac{c}{2}\right)^k$ when $x \neq \frac{c}{2}$; for $0 < k < 1$; $x^k + (c-x)^k < 2\left(\frac{c}{2}\right)^k$ for $k > 1$; $x^k + (c-x)^k > 2\left(\frac{c}{2}\right)^k$ Let $c = a + b$ where $a > 0$, $b > 0$ and $a \neq b$ Thus $0 < a < c$ Case 1: $0 < k < 1$; $a^k + (c-a)^k < 2\left(\frac{c}{2}\right)^k$ $a^k + (a+b-a)^k < 2\left(\frac{a+b}{2}\right)^k$ $a^k + b^k < 2\left(\frac{a+b}{2}\right)^k$ $\frac{a^k + b^k}{2} < \left(\frac{a+b}{2}\right)^k$ Case 2: If $k > 1$; $x^k + (c-x)^k > 2\left(\frac{c}{2}\right)^k$ $a^k + b^k > 2\left(\frac{a+b}{2}\right)^k$ $\frac{a^k + b^k}{2} > \left(\frac{a+b}{2}\right)^k$	3	3Mk: Provides correct solution. 2Mk: Uses part (ii) to show that either $x^k + (c-x)^k > 2\left(\frac{c}{2}\right)^k$ or $x^k + (c-x)^k < 2\left(\frac{c}{2}\right)^k$ depending on the value of k. 1Mk: Finds the value of y when $x = \frac{c}{2}$.
	End of Paper		